

9. Refer to Exercise 4 (Section 15.2). The data in this table are from a paired-difference experiment with $n = 7$ pairs of observations.

Pair	Population						
	1	2	3	4	5	6	7
1	6.9	8.1	9.3	7.7	10.4	8.3	7.4
2	8.8	7.4	9.5	7.8	9.9	8.1	6.9

- a. Use Wilcoxon's signed-rank test to determine whether there is a significant difference between the two populations.

b. Compare the results of part a with the results you got in Exercise 4 (Section 15.2). Are they the same? Explain.

a) H_0 : median difference = 0

H_a : median difference $\neq 0$

Pair	d	rank	signed rank
1	+0.1	1.5	+1.5
2	+0.7	7	+7
3	+0.3	4	+4
4	-0.1	1.5	-1.5
5	+0.5	5.5	+5.5
6	-0.2	3	-3
7	+0.5	5.5	+5.5

$$T^+ = 1.5 + 7 + 4 + 5.5 = 18.5$$

$$T^- = 26.5$$

$$T = 1.5$$

$$T = \min(T^+, T^-) = \min(18.5, 1.5) = 1.5$$

$$T_c = 2$$

$$R = T/2 \rightarrow 1.5 \leq 2 \text{ (Reject } H_0)$$

There is sufficient evidence to conclude that the two populations differ significantly

b) Result is not the same with (16.2) exercise 4

Exercise 4, we failed to reject H_0 , but Wilcoxon rejects H_0 .

Results are different because Wilcoxon use both sign of differences
• magnitude of differences

Wilcoxon is more sensitive and powerful

11. Machine Breakdowns The number of machine breakdowns per month was recorded for 9 months on two identical machines, A and B, used to make wire rope:

Month	A	B	d	d	Rank
1	3	7	-4	4	4
2	14	12	2	2	2.5
3	7	9	-2	2	2.5
4	10	15	-5	5	5
5	9	12	-3	3	3
6	6	6	0	0	0
7	13	12	1	1	1.5
8	6	5	1	1	1.5
9	7	13	-6	6	6

- a. Do the data provide sufficient evidence to indicate a difference in the monthly breakdown rates for the two machines? Test by using a value of α near .05.
b. Can you think of a reason the breakdown rates for the two machines might vary from month to month?

a) H_0 : median difference = 0

H_a : median difference $\neq 0$

$$T = 2.5 + 1.5 + 1.5 = 5.5$$

$$T^- = 2.5 + 5 + 5 + 8 = 20.5$$

$$T = \min(T^+, T^-) = \min(5.5, 20.5) = 5.5$$

$$T_c = 4, n = 9, \alpha = 0.05$$

$$R = T/5 \rightarrow 6.6 \geq 4 \text{ (Failed to reject } H_0)$$

There is not enough evidence to conclude that the two machines have different monthly breakdown rate

b) Number of breakdowns may vary month to month because of workload differences

• maintenance schedules
• operating conditions
• temperature / environment
• variation in production intensity

15. Images and Word Recall A psychology class performed an experiment to determine whether a recall score in which instructions to form images of 25 words were given differs from an initial recall score for which no imagery instructions were given. Twenty students participated in the experiment with the results listed in the table.

Student	With Imagery		Without Imagery	
	With Imagery	Without Imagery	With Imagery	Without Imagery
1	20	5	11	17
2	24	9	12	20
3	20	5	11	20
4	18	9	14	16
5	22	6	15	24
6	19	11	18	22
7	20	8	17	25
8	19	11	18	21
9	17	7	19	19
10	21	9	20	23

- a. What three testing procedures can be used to test for differences in the distribution of recall scores with and without imagery? What assumptions are required for the parametric procedure? Do these data satisfy these assumptions?
b. Use both the sign test and the Wilcoxon signed-rank test to test for differences in the distributions of recall scores under these two conditions.
c. Compare the results of the tests in part b. Are the conclusions the same? If not, why not?

b) SIGN TEST

$$H_0: \mu = 0.5$$

$$H_a: \mu \neq 0.5$$

$$n = 20, x = 20$$

$$X \sim \text{Bin}(20, 0.5)$$

$$P(X \leq 5) = 0.0107$$

$$p\text{-value} = 2(0.0107)$$

$$R = X \leq 5 \text{ or } X \geq 15$$

$$X = 20 \geq 15 \text{ (reject } H_0)$$

WILCOXON

d	rank	d	rank
4	2	10	11
4	2	10	11
4	2	10	11
7	4.5	12	13.5
7	4.5	13	15
6	4.5	15	16.5
6	4.5	15	16.5
9	8.5	16	19
9	8.5	17	20

$$T = 210$$

$$T^- = 0$$

$$T = \min(T^+, T^-) = \min(210, 0) = 0$$

$$n = 20, \alpha = 0.05$$

$$T = 0 \text{ (reject } H_0)$$

c) Both test gave same conclusion : reject H_0

There is sufficient evidence that recall scores differ under the two conditions

6. Legos[®] The time required for kindergarten children to assemble a specific Lego creation was measured for children who had been instructed for four different lengths of time. Five children were randomly assigned to each instructional group. The length of time (in minutes) to assemble the Lego creation was recorded for each child in the experiment.

Training Period (hours)				
5	1.0	1.5	2.0	
8	9	4	4	
14	7	6	7	
19	5	7	5	
12	8	8	5	
10	9	6	4	

Use the Kruskal-Wallis H -test to determine whether there is a difference in the distribution of times for the four different lengths of instructional time. Use $\alpha = 0.1$.

8. Heart Rate and Exercise In Exercise 18 (Section 11.3), we presented data on the heart rates for samples of 10 men randomly selected from each of four age groups. Each man walked a treadmill at a fixed grade for a period of 12 minutes, and the increase in heart rate (the difference before and after exercise) was recorded (in beats per minute). The data are shown in the table.

Training Period (hours)				
5	1.0	1.5	2.0	
8	9	4	4	
14	7	6	7	
19	5	7	5	
12	8	8	5	
10	9	6	4	

- a. Do the data present sufficient evidence to indicate differences in location for at least two of the four age groups? Test using the Kruskal-Wallis F -test with $\alpha = 0.1$.
b. Find the approximate p -value for the test in part a.
c. Since the F -test in Chapter 11 and the H -test are both tests to detect differences in location of the four heart-rate populations, how do the test results compare? If the ANOVA F -test produced the test statistic $F = 0.007$ with $p\text{-value} = 0.988$, compare this p -value with the p -value in part b and explain the implications of the comparison.

H_0 : The distributions of assembly times are the same for all four training periods

H_a : At least one distribution is different

Training Period (hours)

5	1.0	1.5	2.0
8 (8.0)	9 (9.0)	4 (4.0)	4 (4.0)
14 (14.0)	7 (7.0)	6 (6.0)	7 (7.0)
19 (19.0)	5 (5.0)	7 (7.0)	5 (5.0)
12 (12.0)	8 (8.0)	8 (8.0)	5 (5.0)
10 (10.0)	9 (9.0)	6 (6.0)	4 (4.0)

$$BC = 60, 90, 74$$

$$H = \frac{12}{N(N+1)} \sum \frac{T_i^2}{n_i} - 3(N+1)$$

$$= \frac{12}{20(21)} \left[\frac{84^2}{5} + \frac{60^2}{5} + \frac{70^2}{5} + \frac{74^2}{5} \right] - 3(21)$$

$$= 12.71$$

$$X^2_{0.01} = 11.345 \rightarrow 12.71 > 11.345 \text{ (Reject } H_0)$$

There is sufficient evidence to conclude that assembly times differ among the 4 instructional periods

a) H_0 : All four age groups have same distributions

H_a : At least one age group differs

$$H = \frac{12}{N(N+1)} \sum \frac{T_i^2}{n_i} - 3(N+1)$$

$$= \frac{12}{40(41)} \left[\frac{274.5^2}{10} + \frac{164^2}{10} + \frac{216.5^2}{10} + \frac{168^2}{10} \right] - 3(41)$$

$$= 2.63$$

$$X^2_{0.01} = 11.345 \rightarrow 2.63 < 11.345 \text{ (Fail to reject } H_0)$$

There is no sufficient evidence to conclude that

at least two age groups differ in heart-rate increase

$$b) H: 2.63, d.f. = 3$$

$$0.10 < p < 0.40, p \approx 0.49$$

$$0.49 > 0.01 \text{ (Fail to reject } H_0)$$

$$c) p = 0.46 > 0.01 \text{ (Failed to reject } H_0)$$

$$0.46 < 0.49 \text{ (p values are similar)}$$

Both test lead to same conclusion : No significant diff among 4 age groups

Implication

The parametric ANOVA test and nonparametric Kruskal-Wallis test agree

Conclusion is fairly reliable as doesn't depend strongly on normality assumption